

SOFTWARE ENGINEERING

LAB 1

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PROBLEM STATEMENT #61 | Sustainability & Green Tech

Household Carbon Footprint & Target Tracker

1. Problem Context & Overview

A green living analytics tool where households log monthly utility consumption (kWh electricity, gas, commute kilometers) to compute \$CO_2\$ equivalents and track reduction milestones.

Target Stakeholders / Actors: Resident, Sustainability Coach

Requirements Table:

Functional requirement(5):

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest household monthly electricity (kWh) and transport (km) logs to compute total monthly carbon equivalent (CO ₂ e) emissions.	High	Pass: Calculated emissions match the specified EPA conversion factor formulas. Fail: Zero or negative emission values are generated.	This is the core functionality for calculating the household monthly carbon footprint.
FR-002	Functional	The system shall allow the Resident to enter and store monthly electricity consumption in kWh and transport distance in km.	High	Pass: Valid electricity and transport values are successfully stored for the selected month and can be retrieved. Fail:	The system requires monthly consumption data as input for carbon-emission calculations.

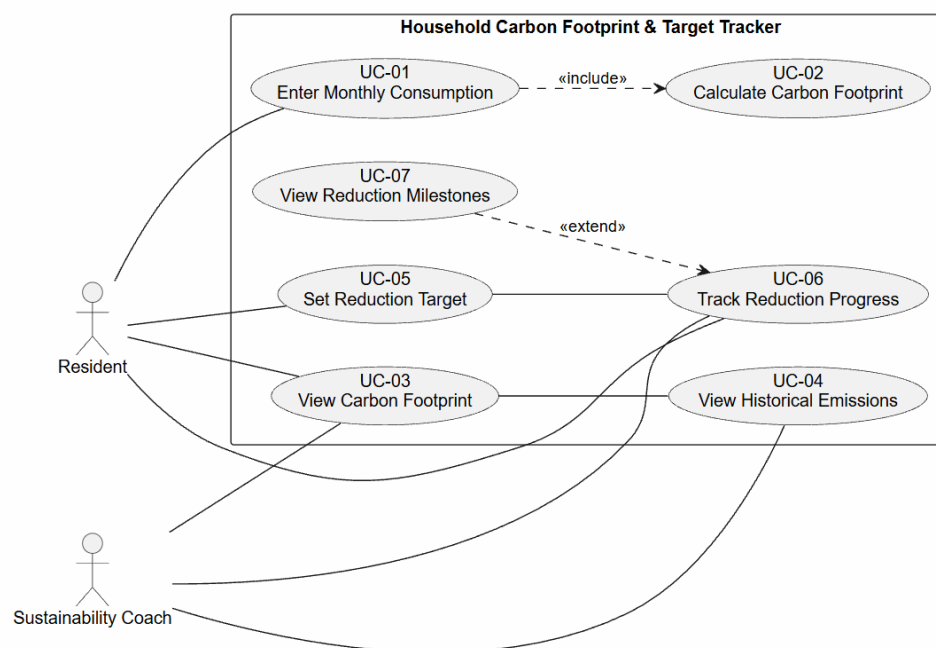
				Missing, invalid, or negative values are stored.	
FR-003	Functional	The system shall display the household's calculated monthly CO ₂ e emissions and historical emission data for comparison.	High	Pass: The system displays the calculated CO ₂ e value for the selected month and previously recorded monthly values. Fail: The calculated value or historical records are missing or incorrect.	Enables the Resident to understand and compare the household's carbon footprint over time.
FR-004	Functional	The system shall allow the Resident to define an annual carbon-reduction target and corresponding reduction milestones.	Medium	Pass: A valid annual reduction target is saved and the corresponding milestones are displayed. Fail: The target is not saved or invalid target values are accepted.	Allows the household to establish measurable carbon-reduction goals.
FR-005	Functional	The system shall track the household's progress toward the defined annual carbon-reduction target and milestones.	High	Pass: After new monthly consumption data is recorded, the system updates the household's progress and correctly indicates achieved milestones. Fail: Progress or milestone status is not updated correctly.	Helps the Resident monitor whether the household is meeting its carbon-reduction goals.

Non-functional requirement(2):

Non-Functional — Performance	The annual carbon reduction milestone chart must render interactive historical comparison graphs in under 200 ms.	High	Pass: Benchmarking tests confirm that the interactive historical comparison graph renders in less than 200 ms under the specified test conditions. Fail: The graph takes 200 ms or longer to render.	Ensures responsive visualization and quick access to historical carbon-footprint comparisons.
Non-Functional — Security	The system shall ensure that household consumption and carbon-footprint data can only be accessed by authorized users.	High	Pass: Authorized users can access the associated household data, while unauthorized access attempts are denied in 100% of security test cases . Fail: Any unauthorized user gains access to household data.	Protects potentially sensitive household consumption and sustainability data from unauthorized access.

UML Use-Case Diagram:

>> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.



- **Use-Case Flow Specification:**

>> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario

>> One Alternate Flow.

Use-Case Flow Specification

UC-01: Enter Monthly Consumption and Calculate Carbon Footprint

Primary Actor: Resident

Supporting Actor: Sustainability Coach

Goal: To record the household's monthly electricity and transport consumption and calculate the corresponding monthly CO₂e emissions.

Preconditions

1. The Resident has access to the Household Carbon Footprint & Target Tracker system.
2. The Resident is authorized to enter household consumption data.
3. The required carbon-emission conversion factors are available in the system.
4. The Resident has the monthly electricity consumption in kWh and transport distance in km.

Postconditions

1. The valid monthly electricity and transport data are stored in the system.
2. The system calculates the household's total monthly CO₂e emissions.
3. The calculated monthly CO₂e value is stored and available for historical comparison.
4. The updated carbon-footprint information can be viewed by the Resident and authorized Sustainability Coach.

Main Success Scenario

1. The Resident selects "**Enter Monthly Consumption**".
2. The system displays a form requesting the household's monthly electricity consumption in **kWh** and transport distance in **km**.
3. The Resident enters the valid electricity consumption and transport distance for the selected month.
4. The Resident submits the consumption data.

5. The system validates that the entered values are present, numeric, and non-negative.
6. The system stores the monthly electricity and transport consumption data.
7. The system applies the configured conversion factors to calculate the household's total monthly **CO₂e emissions**.
8. The system verifies that the calculated emissions are a valid non-negative value.
9. The system stores the calculated monthly CO₂e value.
10. The system displays the calculated monthly carbon footprint to the Resident.
11. The use case ends successfully.

Alternate Flow — Invalid Consumption Data

5a. Invalid or Negative Input

1. At Step 5, the system detects that one or more entered values are missing, non-numeric, or negative.
2. The system displays an error message identifying the invalid input.
3. The system does not store the invalid data or calculate the carbon footprint.
4. The Resident corrects the input values.
5. The system validates the corrected values.
6. The flow resumes from Step 6 of the Main Success Scenario.

Alternate Flow — Calculation Error

7a. Invalid CO₂e Calculation

1. At Step 8, the system detects that the calculated CO₂e value is zero or negative.
2. The system displays a calculation-error message.
3. The system does not save the invalid carbon-footprint result.
4. The Resident is prompted to verify the entered consumption data.
5. The use case ends without completing the carbon-footprint update.